Behzad Boroomandisorkhabi

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Residency Status: US Permanent Resident (Green Card Holder)

Research and Work Experience

Skyworks Solution **Jan. 2025 – June 2025**

RF Test Engineer Co-op

- Conducted EVM (Error Vector Magnitude) testing to assess signal quality and performed DEVM (Differential Error Vector Magnitude) testing to evaluate modulation accuracy.
- Performed AMAM/AMPM tests to evaluate linearity of RF power amplifiers.
- Executed Gain versus Time (GvT) testing to monitor performance over time.
- Tested switching speed to ensure fast and accurate transitions in RF circuits.
- Measured noise figure to evaluate the sensitivity of RF components.
- Conducted small signal tests to analyze low-level signal behavior in RF systems.
- Performed ruggedness tests to validate component durability under extreme conditions.

Missouri University of Science & Technology, Rolla, MO **June 2019 – Aug. 2025**

Research and Teaching Assistant

Research:

- Defined and investigated the electromagnetic properties of materials and measured their dielectric constant by developing one port measurements, simulations, and coding.
- Designed and investigated an active microwave thermography technique to detect covered surface cracks in metal structures.
- Designed and built an ultrafast Lidar sensor by using dispersive Fourier transform to measure instantaneous/microwave frequency and motion with resolution of MHz and um, respectively.
- Developed a time-stretched Lidar sensor with direct optical/microwave processing transduction.
- Created a time-stretched molecular spectroscopy system for the investigation and characterization of gases and liquids.

Taught Courses:

- ▪ Circuit Analysis Laboratory, Control Systems Laboratory, Electronics I Laboratory, Electronic Devices Laboratory and Digital Communication Laboratory.

Coursework:

- Signal Integrity in High-Speed Digital & Mixed Signal Design
- Interference Control in Electronic Systems
- Antennas and Propagation
- Microwave and Millimeter Wave Engineering and Design
- Advanced Electromagnetics

Naghsh Avval Keifiyat (NAK) Company, Tehran, Iran **Aug. 2014 - May 2019**

Senior RF Optimization and Planning Engineer

- Conducted RF optimization for 2G and 3G networks by analyzing network performance and implementing new solutions to improve Key Performance Indicators (KPIs).
- Performed radio network degradation analysis by identifying and analyzing degradation causes and implementing appropriate solutions.
- Executed RF planning for 2G and 3G networks by defining accurate azimuths for new sites to enhance coverage, resolving coverage gaps, estimating required cell capacity, and selecting suitable frequencies to achieve optimal quality and minimize interference.
- Supervised IBS planning and implementation by investigating buildings with coverage gaps, designing IBS plans that included defining antenna positions, measuring link budgets, and estimating required equipment.

Laboratory of Radar Systems, KNTU, Tehran, Iran **Sep. 2011 - Sep. 2013**

Research Assistant

- Designed and fabricated a novel band-notched circular CPW antenna for ultrawideband (UWB) communication in WiFi and WiMAX applications.
- Reduced mutual coupling of SIW slot array antenna and waveguide- fed slot array antenna.

Education

Missouri University of Science & Technology, Rolla, MO **June 2019 – Aug. 2025**
PhD in Electrical Engineering
Coursework: Interference Control, Antenna Propagation, Signal Integrity, Microwave and Millimeter Wave, Advanced Theory of Electric Machines, Advanced Electromagnetic, Advanced Writing for Science and Engineering, Advanced Optical Materials, Optimization under uncertainty.

K. N. Toosi University of Technology, Tehran, Iran **Sep. 2011 - Sep. 2013**
Master of Science in Electrical Engineering – Telecommunications (Fields and Waves)
Thesis: UWB Multi-Band Antennas Design and Investigation Based on Microstrip CPW Antennas
Coursework: Advanced Engineering Mathematics, Electromagnetic Theory, Microstrip Antenna Design, Antenna II, Radio Networks Systems, Radar Systems Design, Quantum Electronics, Active Circuits Design

Laboratory of Wireless Communications, University of Tabriz, Tabriz, Iran **Sep. 2002 - Sep. 2008**
Bachelor of Science in Electrical Engineering – Telecommunications
Thesis: ASK Modulation Simulation

Skills

Devices:
- In depth Practical Experience in working with Spectrum Analyzer/Optical Spectrum Analyzer, Network Analyzer, Optical devices, and components such as laser source, EDFA, Raman laser, splitter, coupler, circulator, polarization controller, polarizer, optical delayer, intensity modulator, collimator, photodetector and balanced photodetector, Oscilloscope (different types)

Telecommunications:
- In depth Practical Experience in Radio Network Optimization and Planning
- In depth knowledge of KPI degradation and root cause analysis
- Practical experience with OSS tools including Business Object, WinFiol, SecureCRT, AMOS, ACTIX, TEMS

Computer Skills:
- **Programming Languages:** MATLAB, C++ and Python
- **Software Packages:** HFSS, CST, COMSOL, Zemax, Actix, TEMS, Atoll, FLEX

Publications

1. Md Abu Zobair, **B. Boroomandisorkhabi** and M. Esmaeelpour, "Recent Advancements in Microwave Frequency Measurement," MDPI, Sensors: review, 25 (12), 2025.
2. **B. Boroomandisorkhabi**, X. Su, and M. Esmaeelpour and, "Picosecond laser ranging at 1.5 microns using dispersive interferometry," Results in Optics, 19, 2025.
3. **B. Boroomandisorkhabi** and M. Esmaeelpour, "Displacement measurement using time-stretch microwave photonics with picosecond laser pulses," Optics and Lasers in Engineering, 186, 2025.
4. **B. Boroomandisorkhabi** and M. Esmaeelpour, "All-fiber real-time ultrafast ranging Lidar for motion management of patients during stereotactic radiotherapy," SPIE Photonics West Conference 2023.
5. **B. Boroomandisorkhabi** and M. Esmaeelpour, "Ultrafast Lidar based on signal time stretch and various transduction techniques," The Magazine of IEEE-Eta Kappa Nu, 2023.
6. **B Boroomandisorkhabi**, S Montazeri, T Kim, M Esmaeelpour, "Head Motion Tracking for Frameless Stereotactic Radiosurgery Using MR-Compatible 1D Lidar," MEDICAL PHYSICS 49 (6), E775-E776
7. **B. Boroomandisorkhabi**, T. Kim, M. Esmaeelpour, "Head Motion Tracking for Stereotactic Radiotherapy Applications," Biophotonics Congress: Biomedical Optics, 2022.
8. F. B. Sorkhabi**, B. Boroomandisorkhabi**, "Directive Pentagon Patch Nano-Antenna for Nanofocusing Application," International Journal of Engineering Research & Technology (IJERT), pp. 769-772, Vol. 9, Issue 09, September-2020.
9. **B**. **Boroomandisorkhabi**, R. Rahimi, "Wide Band E-Shaped Nano-Antenna with Asymmetric Hybrid Plasmonic Waveguide," International Journal of Engineering Research & Technology (IJERT), pp. 391-394, Vol. 9, Issue 07, July-2020.
10. E. Ghahramani, R.A. Sadeghzadeh, **B. Boroomandisorkhabi**, M. Karami, "Reducing mutual coupling of SIW slot array antenna using uniplanar compact EBG (UC-EBG) structure," 8th European Conference on Antennas and Propagation (EuCAP), pp. 2002-2004, 2014.
11. **B. Boroomandisorkhabi**, R.A. Sadeghzadeh, F.B. Zarrabi, and E. Ghahramani, "A Novel UWB Circular CPW Antenna with Triple Notch Band Characteristics," Loughborough Antennas and Propagation Conference (LAPC), pp. 637-640. IEEE, 2013.
12. E. Ghahramani, R.A. Sadeghzadeh, **B. Boroomandisorkhabi**, M.H. Kolagari, "Mutual Coupling Reduction in Waveguide-Fed Slot Array Antenna Using Uniplanar Compact EBG (UCEBG) Structure," LAPC, pp. 634-636, IEEE, 2013.